

Clean Primitive Families and Length Collisions in the Square Billiard

Route-A structural certificate C147

Abstract

Unfolding the unit-square billiard gives an exact census of primitive rational directions and their one-parameter periodic families. The cutoff-40 ledger contains 979 positive coprime ordered directions. The first length collision not induced by coordinate swap is $1^2 + 8^2 = 4^2 + 7^2 = 65$, while an irrational- aspect rectangle removes all such collisions. Every family has unit transverse multiplier, making the ordinary isolated-orbit denominator singular. The natural Dirichlet quantization does not provide a target match.

1 Primitive unfolding

Reflect the unit square across its sides. A regular billiard trajectory becomes a straight line on the doubled square torus. We identify time reversal and label a positive direction by an ordered pair (m, n) with $m, n > 0$. Return of position and oriented velocity requires unfolded displacement $(2m, 2n)$, so

$$L_{m,n} = 2\sqrt{m^2 + n^2}. \quad (1)$$

This return is primitive exactly when $\gcd(m, n) = 1$: a common divisor produces a shorter even translation, while any shorter oriented return produces a common divisor. Vertex-hitting offsets are excluded.

The line crosses $2m$ vertical and $2n$ horizontal sides. Thus its Dirichlet reflection phase is $(-1)^{2(m+n)} = +1$. The horizontal and vertical axis families are boundary classes treated separately from the positive ledger.

2 Periodic cylinders and stability obstruction

For a fixed primitive direction, the transverse offset gives a positive- measure one-parameter cylinder of parallel periodic trajectories. The primitive return preserves this offset. Its reduced Poincaré multiplier is therefore one and

$$\det(I - P_{m,n}) = 1 - 1 = 0. \quad (2)$$

Consequently these are clean families rather than isolated periodic orbits, and the ordinary isolated-orbit determinant convention is unavailable.

3 Exact census and length collisions

Möbius inversion gives the number of ordered positive coprime pairs in the $M \times M$ box:

$$C(M) = \sum_{d=1}^M \mu(d) \left\lfloor \frac{M}{d} \right\rfloor^2. \quad (3)$$

At $M = 40$, this is $C(40) = 979$. Distinguishing time reversal would double the count to 1,958.

The ordered ledger retains coordinate swaps because they are distinct directions, while the nontrivial-collision test quotients that square symmetry. Exact exhaustion finds the first two inequivalent primitive representations at

$$65 = 1^2 + 8^2 = 4^2 + 7^2, \quad (4)$$

giving the common length $2\sqrt{65}$. Every smaller square has at most one primitive representative after sorting the coordinates.

As a geometric control, take a rectangle of width one and height $\alpha = 2^{1/4}$. Squared lengths divided by four are $m^2 + \sqrt{2}n^2$. If two such values agree, rational independence of $1, \sqrt{2}$ forces equality of both integer squares and hence of the positive ordered pairs. The irrational aspect removes every distinct-direction collision, not only those below the ledger cutoff.

4 Quantization, validation, and boundary

The Dirichlet Laplacian on the same square, with domain $H^2(Q) \cap H_0^1(Q)$, is an intrinsic integrable quantization. It neither isolates the cylinders in (2) nor supplies any target comparison.

The independent checker passes 1,079 assertions, SymPy passes 88 checks, byte replay passes, and all 34 hostile receipts (33 repaired-hash and one stale-hash) are rejected. The strict tuple is (A1_WEAK, A2_FAIL, A3_FAIL, A4_NATURAL_QUANTIZATION) overall ROUTE_A_EXPLORATORY. We claim no isolated primitive-orbit determinant, target divisor, functional equation or counting law, prime-like map, arithmetic/local factor, Euler factor, root number, automorphy, Hilbert–Pólya construction, or Route-B authorization. The scope literal is NO_BAD_EULER_OR_ROOT_NUMBER.